Overview

This project attempts to find a solution to Laplace equation $\nabla^2\Psi(r) = 0$ for a given electrode geometry. In particular, we are interested in quadrapole geometry used in electron microscopy or linear paul traps.
Project uses established numerical methods to find such a solution. In particular, we use Finite Elements Methods (FEM) or Boundary Elements Methods (BEM) to find the physical solution, and then we interpolate the result using solid harmonics basis.

The following text describes the technical details of each step.

2. The Boundary Element Method

2.1 The Volume Integral Formulation

From multivariable analysis and potential theory, calculating the scalar potential $\Phi(x, y, z)$ at any observation point $\vec{r}$ due to a continuous internal source distribution (such as charge density or mass) involves integrating over the entire 3D volume V. This is mathematically expressed as the volume integral:

$$\Phi(x, y, z) = \iiint_V \frac{g(x', y', z')}{|\vec{r} - \vec{r}'|} \, dx'dy'dz'$$

In this formulation, $g(x', y', z')$ is the function describing the source density at a given source point $\vec{r}'$. The term $\frac{1}{|\vec{r}-\vec{r}'|}$ represents the distance from the source point to the observation point, acting as the integral kernel that defines how the field's influence decays over space.

While this direct volume integration perfectly describes the physics, evaluating a 3D integral over complex geometries is computationally expensive and requires discretizing the entire internal space.

2.2 Transformation to a Boundary Problem

The Boundary Element Method (BEM) fundamentally transforms this volume-based problem into a surface-based integral problem. For systems in equilibrium where the internal source distribution is zero (governed by Laplace's equation, $\nabla^2\Phi = 0$), the internal state is dictated

entirely by the conditions on the enclosing 2D boundary surface S.

BEM utilizes Green's Identities (in particular, second identity) to push all calculations to this boundary. By introducing a fundamental solution $u^*(\vec{r}, \vec{r}') = \frac{1}{4\pi|\vec{r}-\vec{r}'|}$, which represents the field generated by a concentrated point source, the volume integral collapses. The potential at any observation point $\vec{r}$ can thus be determined entirely by integrating the boundary conditions—specifically the potential u and its normal flux $q = \frac{\partial u}{\partial n}$—over the surface S with respect to the source points $\vec{r}'$.

We introduce the fundamental solution $u^*(\vec{r}, \vec{r}')$, because the electric field arises only where there is potential, and influences only charged particles; since no charged particles are in the vacuum (direct consequence of Laplace equation $\nabla^2 V(\vec{r}) = 0$), the only potential that can destabilize free charged particles on the boundaries of the electrodes, is the boundary itself. Meaning we must find how boundary charges interact with boundary charges through the shared potential field they create. We do not know the phase transition (transient state when charges reallocate to satisfy constant voltage on boundary), only that in the end, they stabilize on a constant voltage.
For simple surfaces such as a sphere, it is possible to exploit symmetries, method of images and other. However, for curved surfaces, or for 2 and more enclosed surfaces, this does not hold, and discretisation (or more complex geometrical transformation) is needed.

2.3 The Boundary Integral Equation

This relationship is mathematically formalized by the Boundary Integral Equation (BIE):

$$c(\vec{r})u(\vec{r}) = \iint_S \left(u^*(\vec{r}, \vec{r}')q(\vec{r}') - q^*(\vec{r}, \vec{r}')u(\vec{r}')\right) dS(\vec{r}')$$

Where:

- $u(\vec{r})$ is the potential at the observation point $\vec{r}$.
- $q(\vec{r}')$ is the normal flux at the source point $\vec{r}'$ on the boundary.
- u^* is the fundamental solution.
- q^* is the normal derivative of the fundamental solution.
- $c(\vec{r})$ is a geometric coefficient ($c = 1$ inside the volume, $c = \frac{1}{2}$ on a smooth boundary).

Because all points evaluated in the integral reside on the boundary, $\vec{r} - \vec{r}' \neq 0$ for almost all interactions. The only exception occurs when the observation point and the source point are

identical ($\vec{r} = \vec{r}'$), which creates a mathematical singularity ($\frac{1}{0}$) that the method must explicitly isolate and resolve numerically.

2.4 Discretization of the Problem

To solve this integral numerically, the global 3D surface S is discretized into non-overlapping elements E_i (e.g., triangles), where each local piecewise function is strictly restricted to the domain $\vec{r} \in E_i$.

To handle arbitrary 3D orientations, the geometry of each element is mapped to a 2D local reference coordinate system (ξ, η) using parametric shape functions N_k, defining any position on the element as $\vec{r}(\xi, \eta) = \sum_k N_k(\xi, \eta)\vec{r}_k$. The unknown physical quantities over the surface are defined using those exact same local shape functions via $w(\vec{r}) = \sum_k N_k(\xi, \eta)w_k$. This isoparametric approach naturally enforces mathematical continuity across shared element boundaries.

2.5 Assembly of the System of Equations

Using the collocation method, we place an observation point $\vec{r}_i$ at a specific node (e.g., the centroid) of each element. The potential $f(\vec{r}_i)$ at element i caused by the physical distribution on element j is calculated by integrating the fundamental solution over the area of element j, yielding a geometric influence coefficient $A_{i,j}$.

According to the principle of superposition, the total potential at element i is the sum of influences from all j elements multiplied by their respective unknown weights w_j. Because the operators are linear, the coefficients $A_{i,j}$ can be isolated into a dense square matrix:

$$\begin{bmatrix} f(\vec{r}_0) \\ f(\vec{r}_1) \\ \vdots \\ f(\vec{r}_i) \end{bmatrix} = \begin{bmatrix} A_{0,0} & A_{0,1} & \cdots & A_{0,j} \\ A_{1,0} & A_{1,1} & \cdots & A_{1,j} \\ \vdots & \vdots & \ddots & \vdots \\ A_{i,0} & A_{i,1} & \cdots & A_{i,j} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_i \end{bmatrix}$$

Expressed in compact vector-matrix notation:

$$\mathbf{F} = \mathbf{A}\mathbf{w}$$

2.6 Transition to BEM Implementation

In a real-world physical problem, boundary conditions are mixed. On some parts of the surface, the potential $\mathbf{F}$ is known and the flux $\mathbf{w}$ is unknown, while on other parts, the reverse is true. Therefore, the system $\mathbf{F} = \mathbf{A}\mathbf{w}$ cannot be solved as originally assembled.

The matrix system must be algebraically rearranged. All columns associated with unknown variables are shifted to a single vector $\mathbf{x}$ on the left-hand side, while all known boundary conditions are multiplied by their respective coefficients and summed into a known vector $\mathbf{b}$ on the right-hand side. This restructuring yields the final linear system:

$$\mathbf{H}\mathbf{x} = \mathbf{b}$$

Because every boundary element geometrically influences every other element, the resulting matrix $\mathbf{H}$ is fully populated, dense, and non-symmetric. The final step is solving this system using standard numerical techniques to yield the complete set of potentials and fluxes across the entire boundary.

3.0 C++ Code Implementation

The theoretical Boundary Element Method is realized in the provided C++ codebase through a collocation-based solver tailored for a multi-electrode physical system (such as a Paul trap). The implementation systematically maps the mathematical formulations into discrete numerical operations using the `Eigen` linear algebra library and the `Gmsh` meshing framework.

Mesh Parsing and Discretization (Section 2.4)

The `load_surface_mesh_from_gmsh` function handles the spatial discretization. It reads a 3D surface mesh and extracts flat 3-node triangular elements (`SurfaceTriangle`). For each triangle, the code calculates the centroid (acting as the collocation point $\vec{r}_i$ where the equations are evaluated), the physical area, and enforces strict outward-pointing normal vectors based on the physical electrode IDs.

Integral Evaluation and Singularity Handling

The geometric influence coefficients for matrices $\mathbf{G}$ and $\mathbf{H}$ are computed in `solve_basis_fields` by evaluating the fundamental solution $\frac{1}{|\vec{r}-\vec{r}'|}$.

- **Smooth Integrals ($i \neq j$):** For off-diagonal elements, the distance $\vec{r} - \vec{r}'$ is strictly non-zero. The integral is solved numerically over the area of element j using a 4-point symmetric Gaussian quadrature rule (`TriangleQuadRule`).
- **Singular Integrals ($i = j$):** When the observation point and source point are on the same triangle, the mathematical singularity is resolved analytically. The `compute_singular_G_ii` function evaluates the weakly singular integral in a closed-form geometric manner. The corresponding singular diagonal for $\mathbf{H}$ is handled via the rigid-body regularization condition (`H(i, i) = -sum(H(i, j))`), guaranteeing exact equipotential conservation.

Matrix Assembly and Solving (Sections 2.5 & 2.6)
Because the physical problem dictates purely Dirichlet boundary conditions—the voltages (potentials) on the electrodes are fixed and known, while the surface charge densities (fluxes) are unknown—the rearranged system $\mathbf{Hx} = \mathbf{b}$ specifically takes the form $\mathbf{Gq} = \mathbf{Hu}$.
The solver constructs a set of 6 basis potential vectors (setting one electrode to 1.0V and all others to 0.0V). It computes the right-hand side $\mathbf{B} = \mathbf{Hu}$ and solves for the unknown surface charges $\mathbf{q}$ using an exact dense LU factorization (`Eigen::PartialPivLU<Eigen::MatrixXd>`).

Superposition and Internal Field Evaluation
Once the basis surface charges are solved, `compute_combined_field` applies the principle of linear superposition to scale the results by any arbitrary configuration of physical voltages.

4.0 Field Interpolation and Solid Harmonic Expansion

While the Boundary Element Method accurately calculates the potential at any internal point, performing a full numerical integration over the entire boundary for every step of a particle trajectory is computationally prohibitive. Furthermore, extracting the electric field ($\vec{E} = -\nabla\Phi$) via finite differences of these discrete points introduces numerical noise and truncation errors.

To resolve this, the code implements a localized analytical interpolation strategy that maps the BEM results to a continuous, perfectly differentiable field within the trapping volume.

The Physical Constraint: Solid Harmonic Basis

Using a standard 3D polynomial fit (like a Taylor series) is dangerous in vacuum electrostatics because arbitrary polynomials do not inherently satisfy Laplace's equation ($\nabla^2 \Phi = 0$). A standard fit could mathematically introduce artificial space charge ($\rho \neq 0$) into the vacuum region.

To guarantee physical accuracy, the `PolynomialInterpolator` class strictly employs a **Solid Harmonic Expansion**. These basis functions are exact, analytical solutions to Laplace's equation in Cartesian coordinates. The implementation defines these terms up to the 4th degree (L=4), yielding a complete basis of 25 spatial terms:

- **Degree 0 (1 term):** Monopole offset.
- **Degree 1 (3 terms):** Uniform uniform dipole fields (x, y, z).
- **Degree 2 (5 terms):** Pure quadrupole fields (e.g., $x^2 - y^2$, $2z^2 - x^2 - y^2$).
- **Degree 3 (7 terms):** Octupole fields.
- **Degree 4 (9 terms):** Dodecapole fields.

Because every basis function in `SolidHarmonicBasis` is an exact solution, their individual analytical Laplacians are identically zero (`laplacian_term`). Consequently, any linear combination of these functions is guaranteed to represent a true vacuum field.

Least-Squares Fitting
The `fit` method takes a dense cloud of internal spatial points (whose potentials were calculated via the BEM solver) and constructs an overdetermined system mapping those coordinates to the harmonic basis. It then solves for the exact multipole coefficients using a robust Column-Pivoting Householder QR factorization
(`A.colPivHouseholderQr().solve(values)`).

Once fitted, the `PolynomialInterpolator` provides continuous $O(1)$ evaluations of the potential and exact analytical gradients for the electric field, eliminating finite difference errors entirely.

5.0 Resolving the round-trip

We fit the result with a harmonic function that literary is a description of quadrapole field. So we perform numerical BEM to find solution, only to fit the solution onto a quadrapole field laplace equation solution. However, there is a real reason why we do this.

1. Real traps aren't perfect hyperbolas.
   A pure, analytical quadrupole field only exists inside a trap built with infinitely large, perfectly hyperbolic electrodes. Traps are built out of cylindrical rods, flat endcaps, or stacked rings. They have gaps, mounting brackets, and finite lengths. It is not possible to write a simple equation for these boundary conditions. The BEM solver is necessary to figure out exactly how these messy, real-world geometries distort the electric field.
2. The higher-order terms are the real prize.
   We aren't just fitting it back to a pure quadrupole. The solid harmonic expansion goes up to the octupole (Degree 3) and dodecapole (Degree 4) terms. The BEM solver figures out the true field, and the harmonic fit acts like a diagnostic tool. It tells us: "Your trap is 98% a perfect quadrupole, but because rods are round instead of hyperbolic, we have a 1.5% dodecapole error and a 0.5% octupole error." Those tiny imperfections (anharmonicities) are exactly what we want to find.
3. Particle tracking hates discrete data.
   If we try to track an ion's trajectory for a million RF cycles using a grid of discrete BEM points, we have to use finite differences to calculate the electric field (the gradient). Finite differences introduce microscopic mathematical noise at every step. Over millions of steps, that noise acts like artificial friction or heating, destroying the simulation.

5.1 Note on interpolation and ordered pairs

The BEM returns us function of potential, stored as a set of ordered pairs of `coordinate:potential`. Mathematically, let position be $\mathbf{r} = (x, y, z)$, with (x, y, z) being the coordinates in the cartesian coordinate basis and u be the potential. If an ordered set of coordinates be $X = \{\mathbf{r}_i \mid i \in \{0, \dots, N\}\}$, and ordered set of potential be $Y = \{u_i \mid i \in \{0, \dots, N\}\}$, then the function of potential $f(X) = Y$ maps linearly all points from potential Y to X.